

Algorithm

Question: 01

01. Express the time complexity of the following function in asymptotic notation.

Tweety (n)

```
{
    j = 1
    while j <= n*n
        j = j * (X MOD 3 + 2)
}
```

How the time complexity will be changed if the **multiplication (*)** is replaced by an **addition(+)**.

Ans:

02. Express the time complexity of the following function in asymptotic notation.

void Wood (int n)

```
{
    for(int i = 1; i <= n; i = i*2)
        for(int j = 1; j <= i; j == j* (Y MOD 3+2))
            k = k + 1;
}
```

How the time complexity will be changed if the **multiplication (*)** is replaced by an **addition(+)**.

Ans:

03. What is the smallest value of n such that an algorithm whose running time is $70*(X+3)*n^2$ runs after faster than an algorithm whose running time is 2^n on the same machine?

Ans:

04. Find the time complexity of the following function?

Fib (n)

```
if (n = 0 or n = 1) return 1
return Fib(n-1) + Fib(n-2)
```

How will it be changed if the function is written as follows?

Fib tail (n, a=0, b=1)

```
if (n == 0)    return a
if (n == 0)    return b
return Fib tail(n-1, b, a+b)
```

Here, a=0, b=1 are default arguments.

Ans:

05. Find the time complexity of the following function.

```
void Tweety (int n)
{
    for(int i = 1; i <= n; i = i++)
        for(int j = 1; j <= i*i; j++)
            k = k + 1;
}
```

How the time complexity will be changed if $i*i$ replaced with i ?

Ans:

06. Find the asymptotic bound of the following recurrence by using master method:

$$T(n) = 4T(n/k) + n^2$$

where (i) $k = 4$ and (ii) $k = 2$.

Ans:

07. Compare the time complexity of the following two functions. Which one runs faster?

```
void Chip (int n)
{
    for(int i = 1; i <= n; i = i++)
        for(int j = 1; j <= i; j = j*2)
            k = k + 1;
}

void Dale (int n)
{
    for(int i = 1; i <= n; i = i*2)
        for(int j = 1; j <= i; j = j++)
            k = k + 1;
}
```

Ans:

08. Let $K = ((X \text{ MOD } 2) + 1) * 2$, where X is the last digit of your ID. Now, find the asymptotic upper bound of the following recurrence by using master method.

$$T(n) = KT(n/2) + n$$

Verify your answer through recursion tree method.

Ans:

09. Show that the asymptotic upper bound of the following function is not $O(n^2)$: $An^3 + Bn^2 + C$. Assume that A, B, and C are any constants.

OR

Show that the asymptotic lower bound of the following function is not $O(n^2)$: $An^3 + Bn^4 + Cn^5$. Assume that A, B, and C are any constants.

Ans:

10. Find the asymptotic upper bound of the following recurrence by using substitution method.

$$T(n) = (2 * (X \text{ MOD } 3) + 2)T(n/4) + n^2$$

Ans: